

📊 PRCP-1003: Customer Transaction Prediction

🎯 Project Overview

Problem Statement

The objective of this project is to analyze customer transaction data and build a predictive model that can identify customers who are likely to perform a transaction in the future.

Business Objective

The bank aims to:

- * Identify customers likely to make future transactions.
- * Improve targeted marketing campaigns.
- * Increase customer engagement.
- * Optimize marketing expenditure.
- * Improve customer retention strategies.

📁 Task 1: Data Analysis Report

Dataset Description

Dataset Information

- * Total Records: ****200,000****
- * Total Features: ****202****
- * Numerical Features: ****200**** (`var\_0` to `var\_199`)
- * ID Column: ****ID_code****
- * Target Column: ****target****

Target Variable

- * ****0**** → Customer will not make a future transaction
- * ****1**** → Customer will make a future transaction

Data Quality Assessment

Missing Values Analysis

☒ No missing values found.

* Total Missing Values = ****0****

Duplicate Records Analysis

☒ No duplicate records found.

* Total Duplicate Records = ****0****

Unique Value Analysis

Observations

* `ID\_code` contains ****200,000 unique customer IDs****

* `target` contains ****2 unique classes****

* All feature columns contain continuous numerical values

Target Variable Analysis

Class Distribution

Class	Count	Percentage
----	-----:	-----:
0	179,902	89.95%
1	20,098	10.05%

Key Observation

⚠ The dataset is highly imbalanced.

* Majority Class (0): 89.95%

- * Minority Class (1): 10.05%

Impact

- * Accuracy alone is not sufficient.

- * Additional metrics are required:

- * Precision

- * Recall

- * F1-Score

- * ROC-AUC

Statistical Summary

Observations

- * Features have different scales.

- * Some features contain negative values.

- * No abnormal data quality issues found.

- * Feature scaling is required for scale-sensitive algorithms.

Data Preprocessing

ID Column Removal

- * Removed `ID\_code`
- * Reason: Customer ID does not contribute to prediction.

Feature Scaling

Applied ****StandardScaler**** to:

- * Logistic Regression
- * K-Nearest Neighbors (KNN)
- * Gaussian Naive Bayes
- * Support Vector Machine (SVM)
- * Artificial Neural Network (ANN)

Not required for:

- * Random Forest
- * XGBoost
- * LightGBM
- * CatBoost

Train-Test Split

Dataset split using ****80:20 ratio****

- * Training Data: 160,000 records

- * Testing Data: 40,000 records

Used:

- * Stratified Sampling

Reason:

- * Preserves class distribution in train and test sets.

Exploratory Data Analysis (EDA)

Key Findings

- * No missing values.
- * No duplicate records.
- * Significant class imbalance exists.
- * Features are anonymized.
- * Numerical features have varying distributions and scales.

📁 Task 2: Predictive Modeling

Objective

Develop predictive models to identify customers likely to make future transactions.

Models Implemented

Machine Learning Models

1. Logistic Regression
2. Random Forest
3. Decision Tree
4. Gaussian Naive Bayes
5. XGBoost
6. LightGBM
7. CatBoost

Deep Learning Model

8. Artificial Neural Network (ANN)

Model Evaluation Metrics

The following metrics were used:

- * Accuracy
- * Precision
- * Recall
- * F1-Score
- * ROC-AUC

Why ROC-AUC?

Since the dataset is imbalanced, ROC-AUC provides a better evaluation of model performance than Accuracy.

📄 Model Comparison Report

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
-----	-----:	-----:	-----:	-----:	-----:
Logistic Regression	0.7925	0.2974	0.7816	0.4309	0.8657
Random Forest	0.8995	0.0000	0.0000	0.0000	0.8097
Decision Tree	0.7262	0.1751	0.4647	0.2543	0.6371
Gaussian Naive Bayes	0.9198	0.7008	0.3525	0.4690	0.8860
XGBoost	0.9156	0.8306	0.2012	0.3240	0.8824
LightGBM	0.9186	0.8096	0.2485	0.3803	**0.8910**
CatBoost	0.9157	**0.8438**	0.1975	0.3201	0.8855
ANN	0.9107	0.6243	0.2799	0.3865	0.8505

Model Ranking (Based on ROC-AUC)

 ****LightGBM**** → 0.8910

 ****Gaussian Naive Bayes**** → 0.8860

 ****CatBoost**** → 0.8855

 XGBoost → 0.8824

 Logistic Regression → 0.8657

 ANN → 0.8505

 Random Forest → 0.8097

 Decision Tree → 0.6371

🔍 Feature Importance Analysis

Feature importance was analyzed using:

* Logistic Regression

- * Random Forest

- * XGBoost

- * LightGBM

- * CatBoost

Top 5 Features Identified by LightGBM

Rank	Feature
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1	var_76
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2	var_174
---	---------

3	var_146
---	---------

4	var_34
---	--------

5	var_6
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Observation

These variables contributed the most to transaction prediction.

Since the dataset is anonymized, business interpretation is not possible.

 Hyperparameter Tuning

Technique Used

* RandomizedSearchCV

Best Parameters

* num_leaves = 50

* n_estimators = 500

* learning_rate = 0.05

Tuned Model Performance

* Accuracy = 0.9181

* Precision = 0.8165

* Recall = 0.2391

* F1-Score = 0.3698

* ROC-AUC = 0.8875

Conclusion

The tuned model did not outperform the original LightGBM model.

Therefore, the original LightGBM model was retained.

 Final Model Selection

Selected Model: LightGBM

Reasons

- * Highest ROC-AUC Score (0.8910)
- * Strong overall predictive performance
- * High Accuracy (91.86%)
- * Good Precision (80.96%)
- * Handles large datasets efficiently
- * Fast training and prediction
- * Suitable for production deployment

🧩 Challenges Faced and Solutions

Challenge 1: Class Imbalance

Problem

- * Class 0 = 89.95%
- * Class 1 = 10.05%

Solution

- * Stratified Train-Test Split
- * Used ROC-AUC, Recall, Precision, and F1-Score

Reason

Accuracy alone is misleading for imbalanced datasets.

Challenge 2: Anonymized Features

Problem

Business meaning of features was unavailable.

Solution

Used:

- * Feature Importance Analysis

- * Statistical Analysis

Reason

To identify influential predictors despite lack of domain information.

Challenge 3: High Dimensionality

Problem

200 numerical features increased complexity.

Solution

- * Feature Importance Analysis
- * Ensemble Models

Reason

Boosting algorithms effectively handle high-dimensional data.

Challenge 4: Model Selection

Problem

Different algorithms produced different performance levels.

Solution

Compared multiple models using common evaluation metrics.

Reason

To identify the most reliable model.

Challenge 5: Hyperparameter Optimization

Problem

Default model settings may not be optimal.

Solution

Applied RandomizedSearchCV.

Reason

Improves model generalization and robustness.

Business Impact

The final model can help the bank:

- * Identify potential future customers.
- * Improve campaign targeting.
- * Increase customer engagement.

- * Optimize marketing budget.
- * Improve customer retention.
- * Support data-driven decision-making.

🔄 Conclusion

A complete end-to-end Machine Learning pipeline was developed for Customer Transaction Prediction.

Key achievements:

- ✓ Data Cleaning and Validation
- ✓ Exploratory Data Analysis
- ✓ Feature Engineering
- ✓ Multiple Model Development
- ✓ Hyperparameter Tuning
- ✓ Model Comparison
- ✓ Feature Importance Analysis

After evaluating multiple Machine Learning and Deep Learning models, ****LightGBM achieved the highest ROC-AUC score (0.8910)**** and demonstrated the best overall predictive performance.

Final Recommendation

 ****Deploy LightGBM as the production model for customer transaction prediction.****